

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

## Chapter 5: Quadratic Equations

Every formula and result of the chapter on one sheet

|    |                                          |                                             |
|----|------------------------------------------|---------------------------------------------|
| 1  | Standard form                            | $ax^2 + bx + c = 0, a \neq 0$               |
| 2  | Zero product rule                        | If $pq = 0$ , then $p = 0$ or $q = 0$       |
| 3  | Quadratic formula                        | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$    |
| 4  | Discriminant                             | $D = b^2 - 4ac$                             |
| 5  | Real and unequal roots                   | $D > 0$                                     |
| 6  | Real and equal roots                     | $D = 0$ ; each root $= -\frac{b}{2a}$       |
| 7  | Roots not real                           | $D < 0$                                     |
| 8  | Real roots                               | $D \geq 0$                                  |
| 9  | Sum of the roots                         | $-\frac{b}{a}$                              |
| 10 | Product of the roots                     | $\frac{c}{a}$                               |
| 11 | Equation with roots $\alpha$ and $\beta$ | $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ |